Tomato Light Guide

How much light tomatoes need, grow lights, and how to fix light problems

Quick answer: Tomatoes are sun-loving plants. They need **at least 6-8 hours of direct sun** a day, and 8-10 hours gives better yield. They are day-neutral, so the day length does not decide when they flower. Indoors, use bright grow lights for **14-16 hours** a day, and give the plants a dark period.

1. Light needs by growth stage

Stage	Light needs	Notes
Germination	Light is not needed for most seeds	Warmth is more important. Give light as soon as seedlings appear.
Seedlings	Bright light, 14-16 hours (indoors)	Weak light makes tall, thin, weak seedlings.
Vegetative growth	Full sun, 8+ hours	Strong light makes thick stems and compact plants.
Flowering and fruit set	Full sun, 8-10 hours	Low light causes flower drop and poor fruit set.
Fruit ripening	Good light on leaves, some shade on fruit	Leaves make the sugar. Too much direct sun on fruit can cause sunscald.

2. Light intensity numbers

Growers measure light in two ways. **PPFD** is the amount of light the plant can use at a moment (umol/m^2/s). **DLI** (daily light integral) is the total amount over one day (mol/m^2/day).

Stage	PPFD (approx.)	DLI (approx.)
Seedlings under grow lights	200 to 400 umol/m^2/s	12 to 20 mol/m^2/day
Growing and fruiting plants	400 to 800 umol/m^2/s	20 to 30 mol/m^2/day
Full summer sun (outdoors, midday)	1500 to 2000 umol/m^2/s	Often 40 or more

The numbers are typical targets. Higher light gives higher yield if temperature, water and nutrients are also good.

3. Signs of wrong light

Problem	Signs	What to do
Too little light	Tall, thin, pale (leggy) plants; yellow lower leaves from shading; few flowers; flower drop; small, less sweet fruit; more mold and disease in humid places	Move to a sunnier place; add or lower grow lights; remove extra leaves and prune for light; increase spacing
Too much light or heat	Bleached or scorched leaves; leaves rolling and curling; sunscald (white or yellow patches on fruit); yellow shoulders on fruit	Use 30-40% shade net in the hottest weeks; keep enough leaves to shade the fruit; water regularly; harden off seedlings first
Light all day and night (24 hours)	Yellow patches between veins and mottled leaves (continuous light injury)	Give at least 6-8 hours of darkness. Use 14-16 hours of light.

Yellow leaves: yellow lower leaves in a crowded plant are often just shaded and old. Yellow, bleached patches on top leaves in strong sun can be sun damage. Check water, nutrients, pests and disease too.

4. Grow lights for indoor tomatoes

- Use **full-spectrum LED** grow lights (white light with red and blue). They use less power and give less heat.
- Keep the lights close at first, usually about 5-15 cm above the seedlings for most LED panels. Follow the maker's advice, because strong lights need more distance.
- Raise the lights as the plants grow.
- Use a timer: about 14-16 hours on and 8-10 hours off.
- Use a fan to stop heat and humidity from building up under the lights.

5. Managing light outdoors and in greenhouses

- Choose the sunniest spot with at least 8 hours of direct sun. Avoid shade from trees and walls.
- **Harden off** seedlings: put them outside a little longer each day for 7-10 days before planting, to avoid leaf burn.
- Plant rows north to south so both sides of the row get sun.
- Leave space between plants, use stakes or trellises, and remove extra shoots so light reaches the leaves and fruit.
- In winter greenhouses, keep the roof clean and add supplemental lighting if daylight is short.
- In hot summers use shade net or white wash on the greenhouse.

6. Simple checklist

- At least 6-8 hours of direct sun, ideally 8-10.
- Indoors: 14-16 hours of full-spectrum light, with a dark period.
- Harden off seedlings before moving them into strong sun.
- Prune and space the plants so light reaches the leaves.
- Use shade in the hottest weeks, and keep enough leaves to cover the fruit.
- Check light together with temperature, water and nutrients.

Note: Light needs depend on the variety, climate and growing system. Light meter apps are not very accurate. Use a quantum (PAR) meter if you need exact numbers.